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Review

Reviewed Work(s): *The Handbook of Research Synthesis* by H. Cooper and L. Hedges

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proliferating collection of mathematically oriented books on computations that generally constitute the textbook literature on artificial intelligence (AI). The publisher lists the book in its Applied Information Technology series. One needs to read Chapter 1, "Statistics and Computing: The Promise and the Risk," to learn that the 12 chapters with separate authors are selections from, and a few additions to, the presentations at a technical conference, "AI and Computer Power: The Impact on Statistics," March 1991, in London. Despite the subtitle, 9 of the 12 chapters emanate from computer sciences departments in France, the United Kingdom, and the United States.

The short introductory chapter discusses the impact of computing on classical statistical techniques, reviews the methods only possible now because of computing power and gives a perspective on statistical expert systems. The second of the statistics chapters deals with GLIMPSE, the expert-advisor front-end for GLIM, and the lessons that were learned from building it. The third statistics chapter is titled "Computationally Intensive Methods in the Design of Experiments." Mostly, this paper deals with *D*-optimal designs and their computations.

Several of the other chapters deal with simulation, which may interest statisticians. The other prominent topic is "reasoning," for which the chapter topics are reasoning under uncertainty, diagnostic reasoning, and practical probabilistic reasoning. Most of the book is descriptive, the chapters generally being overviews rather than developments of methodology.

A Handbook of Small Data Sets, edited by D. HAND, F. DALY, A. LUNN, K. MCCONWAY, and E. OSTROWSKI, London: Chapman and Hall, 1994, xi + 458 pp. \$64.95 w/disk.

This is the second book of data sets that has been published in the past few years. See Ziegel (1987) for a report on Andrews and Herzberg (1985). There are a couple of significant differences between this book and the predecessor. The earlier book had only 100 data sets, whereas this one has 500! Here they are all "small," whereas the previous book had large ones too. Most important though, is that this new book comes with a data disk that contains all 500 data sets. Any of us at some time needs quickly to have a certain type of data set. Here is a great source.

A tabular listing of each data set composes much of the book. There is more, however. First there is a citation for the source of the data, such as John (1961) for data number 13, "Comparing Dishwashing Detergents." There also is a paragraph that describes the problem and the collection of the data. Here the editors suggest the kind of analysis for which the data can be used. Generally this stipulates the analysis of the citation, but sometimes alternatives are suggested.

There are two indexes at the back of the book. First, there is the Data Structure Index. For all 500 sets, in the numerical order of the book, three pieces of information are given. The data structure gives two numbers, the number of data points and a second number that I did not understand. The categorization gives the types of the variables. Nominal, numeric, binary categorical, rate, and count are some of the listings. Finally, the filename is given, such as DISHWASH.DAT. The second index is a subject index.

Curiously there seems to be no order to the data sets themselves. I would have created chapters for the most likely type of statistical analysis. In addition, there is no index by type of statistical analysis, which is not very helpful, for example, if you need a nice illustration for three-way analysis of variance. If you have need of some data, though, ready to analyze, this may be the source for you.

REFERENCES

- Andrews, D., and Herzberg, A. (1985), *Data*, New York, Springer-Verlag.
John, P. W. M. (1961), "An Application of a Balanced Incomplete Block Design," *Technometrics*, 3, 51–54.

Ziegel, E. (1987), Review of *Data*, by D. Andrews and A. Herzberg, *Technometrics*, 29, 502.

The Handbook of Research Synthesis, edited by H. COOPER and L. HEDGES, New York: Russell Sage Foundation, 1994, xvi + 573 pp., \$49.95.

Is research synthesis at all relevant in the physical and engineering sciences? Those of us in the petroleum refining and marketing business know about the USEPA's complex model, a gasoline emission model synthesized from data from several oil companies. The editors in the foreword mention only information scientists, psychologists, social policy analysts, and educators as practitioners who might use meta-analysis and have contributed papers to this volume. Medical research was the application area most familiar to me as using meta-analysis. Some contributors here also were statisticians.

This book is the third in a series on research synthesis by this publisher. The book has 32 chapters partitioned into 10 sections. According to the foreword, each of the sections deals with a "central decision point" in the conducting of a research synthesis project. Each chapter was intended to involve examining an "issue" within each decision point. The principal decision points are problem formulation, literature searching, literature evaluation, statistical description and combination of studies, statistical analysis of the size of an effect, and reporting the results. Some special topics are also covered.

The editors gave considerable attention to details, as reported also in the foreword. A consistent mathematical notation, a common use of terminology, and the rounding of numerical results were issues that were mentioned. The book is printed densely in double columns and surely is the most comprehensive treatment that can be imagined for this topic.

The Collected Works of John Tukey: Volume VIII, Multiple Comparisons, edited by H. BRAUN, New York: Chapman and Hall, 1994, lxi + 475 pp., \$62.95.

This is the eighth volume in a monumental collection of publications. The papers here cover 1948–1983 and deal with a topic widely associated with John Tukey because one of the well-known tests and philosophies for multiple comparisons bears his name. Most of us probably were not aware that 11 publications over 35 years were written by Tukey on this topic.

The book begins with the same introductory material that appears in the other seven volumes. [For details, see Ziegel (1988)]. Here nine introductory pages, a foreword, and the editor's comments on the papers in this volume, a feature that is one of the highlights of the series, constitute the unique material. About two-thirds on this volume comprises a 300-page, previously unpublished Princeton University manuscript, "The Problem of Multiple Comparisons." Another 100 pages are a *Technometrics* paper (Tukey 1965) and a *Biometrika* paper (Kurtz, Link, Tukey, and Wallace 1965), on "short-cut multiple comparisons." The remainder of the pages contain mostly notes, comments, and author replies, generally previously unpublished.

The book is identically bound to previous volumes and reasonably priced but probably limited in audience to collectors, libraries, and persons doing research work in statistical comparisons.

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- Kurtz, T., Link, R., Tukey, J., and Wallace, D. (1965), "Short-Cut Multiple Comparisons for Balanced Single and Double Classifications: Part 2, Derivations and Approximations," *Biometrika*, 52, 485–498.
Tukey, J. (1965), "Short-Cut Multiple Comparisons for Balanced Single and Double Classifications: Part 1, Results," *Technometrics*, 7, 95–161.
Ziegel, E. (1988), Review of *The Collected Works of John Tukey: Volume 3*, edited by L. Jones, *Technometrics*, 30, 363.